Supplementary Information for "Mesoscale air motion and thermodynamics predict heavy hourly U.S. precipitation"

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Supplementary Notes 1.

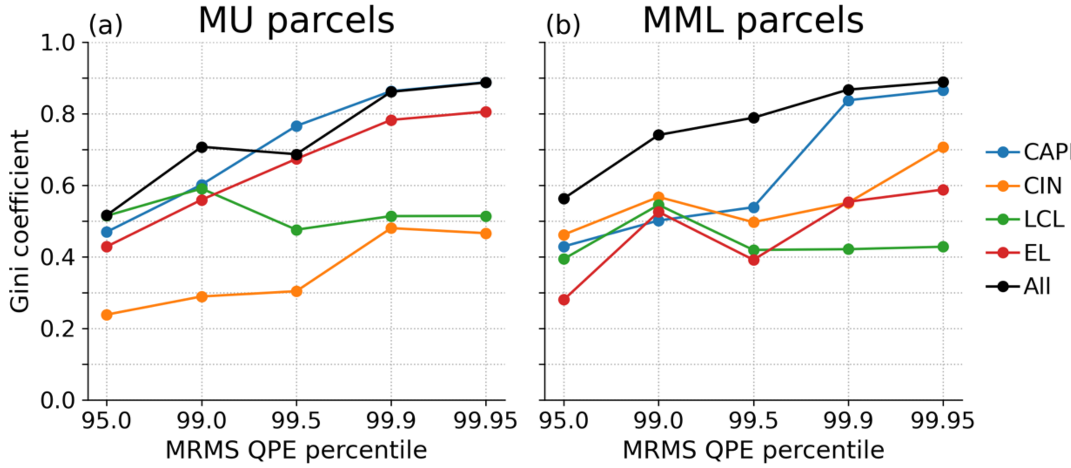
CE-CONUS Gini coefficients were calculated for QPE>QPE_X thresholds with each of four convective indices: convective available potential energy (CAPE), convective inhibition (CIN), lifted condensation level (LCL) and equilibrium level (EL). Both most unstable (MU) and mean-mixed layer (MML) parcels were used. Indices were ranked either in order (CAPE, EL) or reverse order (CIN, LCL) based on physical expectations. Predictions were also made using combinations of indices, for example fitting an ordinary least squares regression to all four predictors:

$$QPE = \beta_4 CAPE + \beta_3 CIN + \beta_2 LCL + \beta_1 EL + \beta_0 \tag{S1}$$

After obtaining sample estimates of the $\hat{\beta}$ regression coefficients from a randomly selected training subset, the convective indices were put into Eq. (S1) and the QPE estimate was treated as the predictor, and the Gini coefficient derived. For individual indices, the result is identical to ranking the predictors alone, since Eq. (S1) would collapse, e.g. for CAPE to $QPE = \beta_4 CAPE + \beta_0$, and the relative ranking of predictors is equal to the ranking by CAPE alone.

Supplementary Figure 1 shows that MU or MML CAPE alone provides almost all the Gini coefficient skill. The most extreme events are predicted more by CAPE than any other convective index alone, and while MU_CAPE and MML_CAPE show similar Gini coefficients for QPE>$QPE_{99.9}$ or above, MU_CAPE shows more consistently high Gini coefficient across the QPE_X thresholds so all main manuscript results use MU_CAPE.

For the parameter estimation of Eq. (S1), we performed the regression using a randomly selected training sample of N=20,000, and Gini coefficients were derived excluding the training sample. The sample size was selected based on bootstrapped sensitivity tests confirming that this sample size resulted in stable Gini coefficients. Performance was also calculated using all 2- and 3-parameter combinations, and using random forest regression in addition to OLS. The conclusions about using MU_CAPE alone were not affected by these tests (not shown).



Supplementary Figure 1: Gini coefficient derived for select MRMS QPE percentile thresholds using either individual convective index predictors, or using a combination of all predictors. (a) using most unstable parcels, (b) using mean-mixed layer parcels.

Supplementary Notes 2.

A common metric used to judge classifier skill is the area under the curve (AUC) of the ROC. The ROC is calculated by plotting the true positive rate (TPR) against the false positive rate (FPR) for a range of predictor thresholds, and then calculating the integrated area under this curve. The TPR is derived from the counts of true positive (TP) and false negative (FN) entries returned by the classifier:

$$TPR = \frac{TP}{TP+FN} \tag{S2}$$

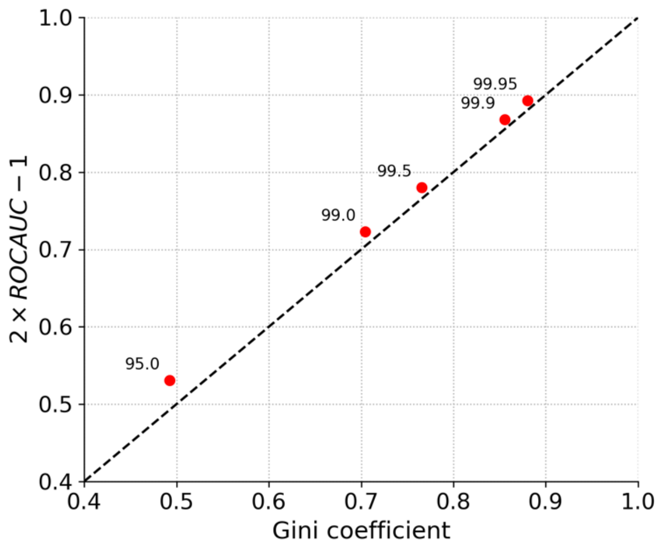
Meanwhile the false positive rate (FPR) is derived from the counts of true negative (TN) and false positive (FP) values:

$$FPR = \frac{TN}{TN+FP} \tag{S3}$$

The main manuscript references discuss pros and cons of the use of the ROC for inferring classifier performance, and note that a common metric derived from the ROC curve is to integrate the area under the curve (AUC). Under certain conditions, there is a relationship between the ROCAUC and the Gini coefficient:

$$Gini \approx 2ROCAUC - 1 \tag{S4}$$

Supplementary Figure 2 shows that Eq. (S4) approximately holds in this case, and that the Gini coefficient we report is closely related to the ROCAUC. We present the Gini coefficient because it is derived directly from the CDFs shown in Figures 2 & 3, which are easily and intuitively interpreted in terms of how many QPE>QPE_X events occur for a given CAPE≤$CAPE_X$ threshold.



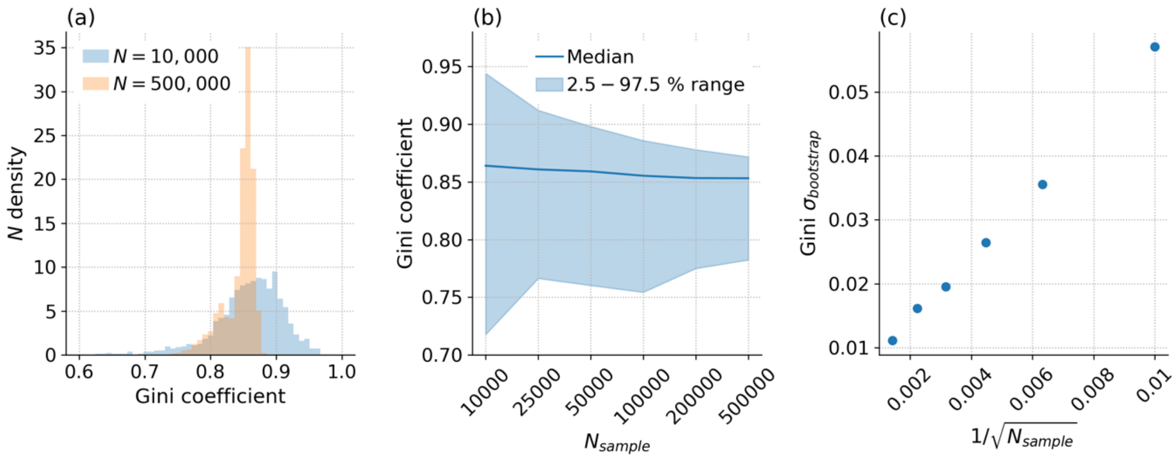
Supplementary Figure 2: Receiver operator characteristic area under the curve (ROCAUC) scaled by 2×ROCAUC-1 as a function of the Gini coefficient derived using MU_CAPE as a predictor and for QPE_X percentile thresholds as labelled in the figure.

Supplementary Notes 3.

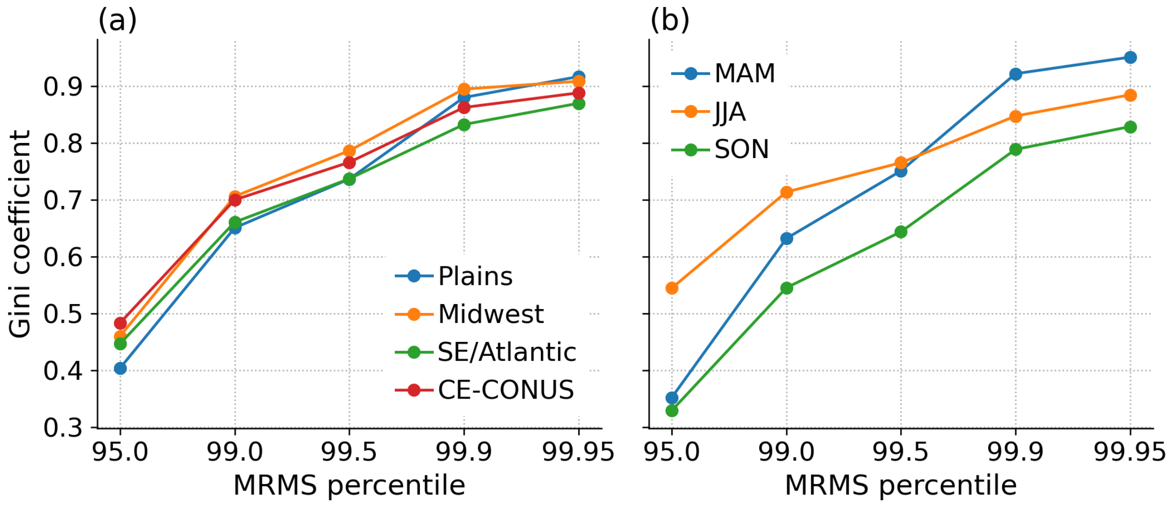
Rare events require large sample sizes to assess so we use derive our thresholds from the full dataset. With *N*>1 million, our 99.95th percentile captures approximately the 600 most-intense events. A major concern of using the full sample is that, rather than identifying locally rare events, we may simply be identifying seasons or places where heavy precipitation is more common. Despite this concern, we use the same space-time samples for all products, so any improvements from AIRS-FCST relative to AIRS overpass, ERA5 or HRRR CAPE represents legitimately improved skill from trajectory enhancement.

It would be preferable to have time- or region-varying thresholds to define more intense events, but with 455 days of valid data, it is impossible to confidently derive rare percentiles such as the 99.95th on the smallest spatial scales. We bootstrap the data to estimate how sample size affects the sampling distribution of the Gini coefficient for $QPE_{99.95}$. Supplementary Figure 3(a,b) show how the sampling distribution is skewed, but narrows as sample size increases. Supplementary Figure 3(c) shows how the magnitude of the standard deviation of the sampling distribution (aka the standard error, σ) scales with $1/\sqrt{N}$, as commonly occurs. While the skewed distribution means that σ doesn't fully capture uncertainty, the scaling with $\sqrt{N}$ in panel (c) allows us to scale the results from (b) to estimate confidence intervals for an arbitrary *N*. In this manner, the 95 % confidence intervals for the full sample is estimated to span approximately [-0.018,0.013]. The bounds are sufficiently tight to support the paper's primary conclusions.

We test the sensitivity of our results to regional and temporal subsetting, subsetting by regions or times to ensure sample size >100,000, to avoid the larger errors associated with smaller samples. Supplementary Figure 4 shows that the derived Gini coefficients vary somewhat when the percentile thresholds are defined for sub-regions (defined in Supplementary Table 1) or seasons. Supplementary Figure 5 performance does vary by season, with AIRS-FCST marginally outperforming even HRRR QPE at $QPE_{99.95}$ during spring and summer, but falling behind HRRR QPE and ERA5 CAPE by autumn. Overall, we do not consider these minor changes in ranking to undermine the paper's primary conclusions.



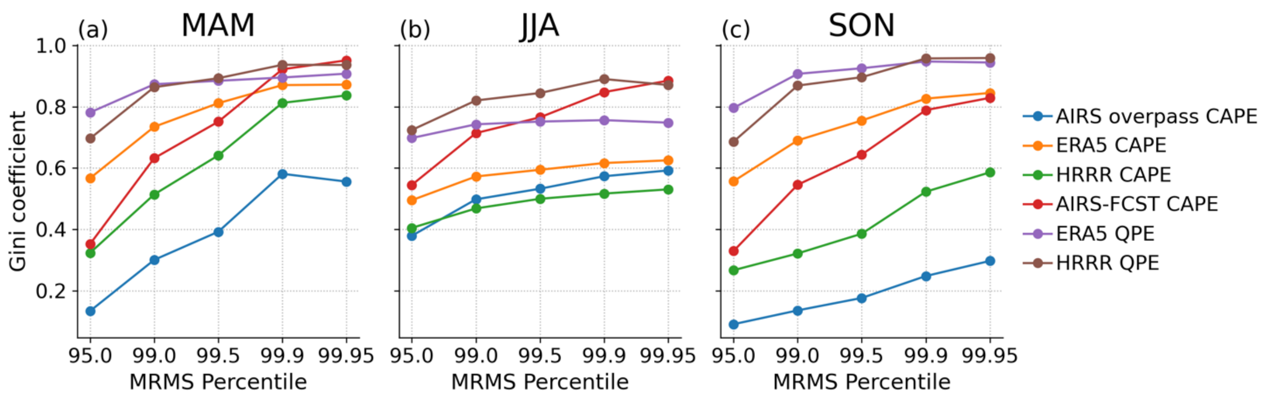
Supplementary Figure 3: Statistics derived from bootstrap resampling of the AIRS-FCST CAPE and MRMS QPE data, then deriving the Gini coefficient for QPE>$QPE_{99.95}$ on each sample. (a) Distribution of the bootstrapped Gini results for *N* of 10,000 or 500,000. (b) Median and 5—95 % range of Gini coefficient distribution by sample size. (c) standard deviation of the bootstrapped samples as a function of $1/\sqrt{N}$.



Supplementary Figure 4: Derived Gini coefficients as a function of the MRMS percentile, with percentiles derived within (a) individual regions as defined in Supplementary Table 1 or (b) over all of CE-CONUS but individual seasons.

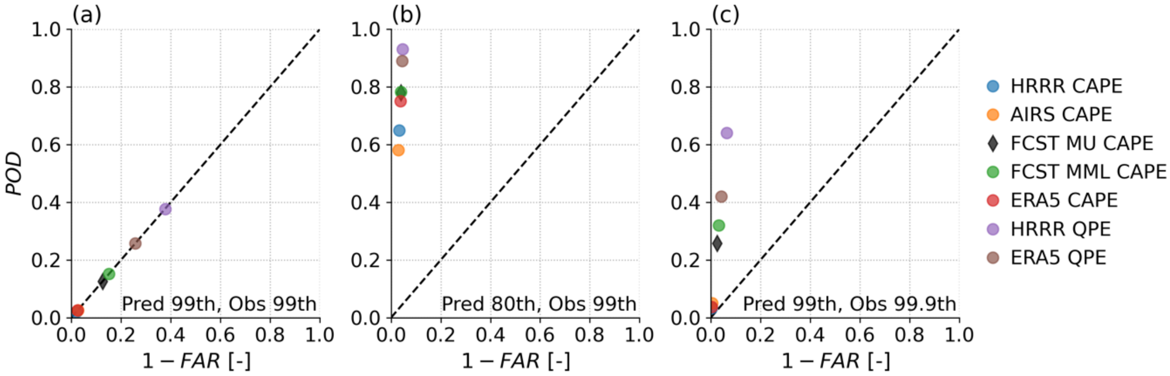
Supplementary Table 1: Region definitions as used in Supplementary Figure 4 and sample size statistics.

	Lat	Lon	N
Plains	32—53 °N	107—96 W	176,352
Midwest	36—53 °N	96—80 °W	384,822
SE/Atlantic	32—36 °N	96—65 °W	337,092

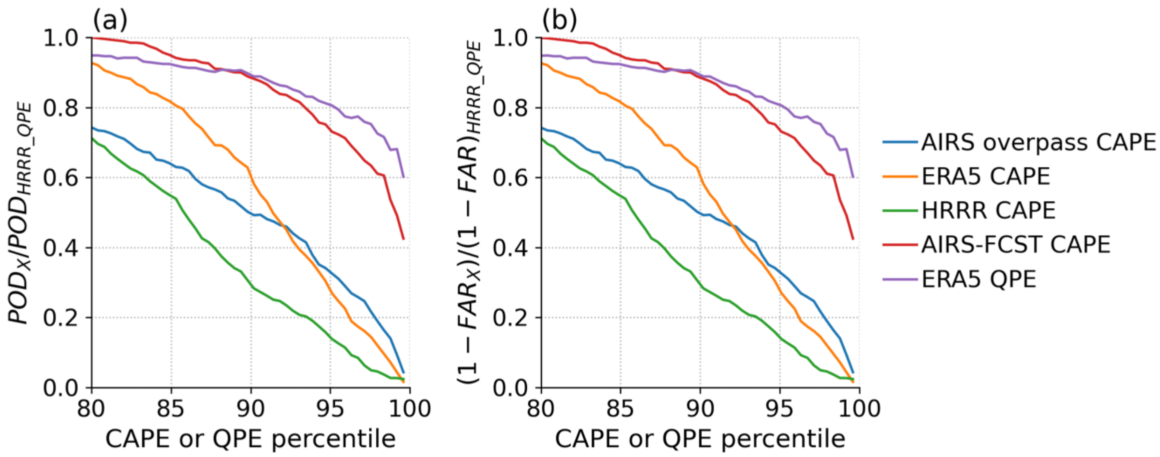


Supplementary Figure 5: Product performance comparison for each season.

Supplementary Notes 4. Other skill score characteristics



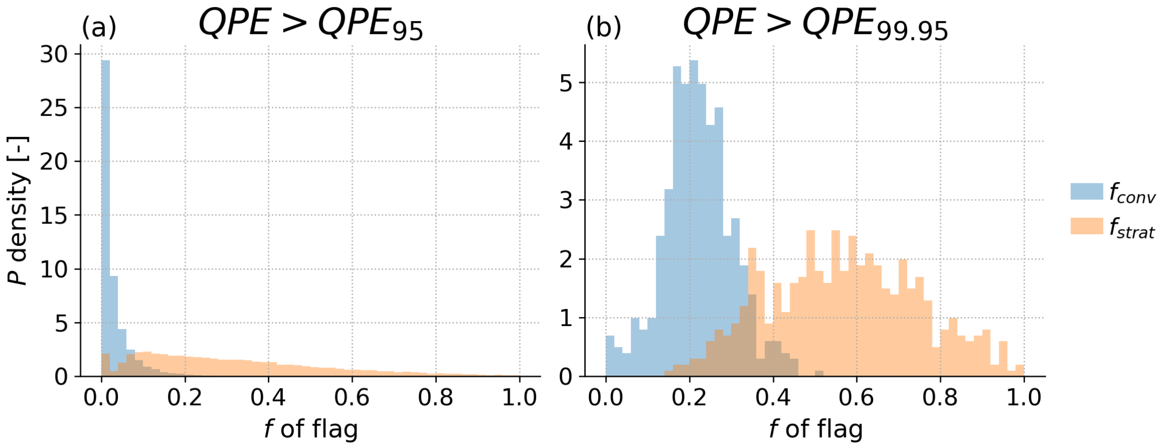
Supplementary Figure 6: Skill diagrams showing the probability of detection (POD) as a function of one minus the false alarm ratio (FAR), shown for select $CAPE_X$ and QPE_X thresholds. Improved skill means movement towards the upper-right corner. The typical ranking follows that reported in the main paper conclusions, with the AIRS-FCST CAPE (shown here are both most-unstable and mean-mixed-layer CAPE) outperforming AIRS overpass, ERA5 or HRRR.



Supplementary Figure 7: For $QPE > QPE_{99.95}$ (a) probability of detection for each product divided by the value derived using HRRR QPE, as a function of the predictor (CAPE or QPE) percentile, (b) 1 minus the false alarm ratio for the same. The HRRR QPE was identified as the best predictor, so a score of one means that at the selected predictor threshold the product equals HRRR QPE. The rankings are largely consistent: AIRS-FCST CAPE is similar in skill score to ERA5 QPE, and exceeds the skill scores of all other products, with substantial improvements relative to the other CAPE values.

Supplementary Notes 5.

MRMS data include a precipitation flag. We cluster hail, convection and tropical convection together as "convective" and warm, cool and tropical stratiform together as "stratiform", then Supplementary Figure 8 shows the histograms of the fractional area coverage of columns identified as each type during events where QPE exceeds its 95th or 99.95th percentile. Note that QPE is calculated as a grid cell mean, so the total area coverage will only equal one where there is total precipitation coverage. As expected, there is a shift to increasing convective occurrence at heavier precipitation levels, and it is also rare that grid cells are entirely convective, since radar products classify profiles, and convective systems are often associated with some regions of stratiform precipitation, such as particles precipitating from anvils. However, approximately 13,000 of the timesteps in the QPE>QPE_X steps have zero convective flags in the associated 1°×1° grid cell, so in the main text we refer to QPE>QPE_{95} as including larger contributions from stratiform or shallow convective precipitation, relative to higher QPE_X thresholds.

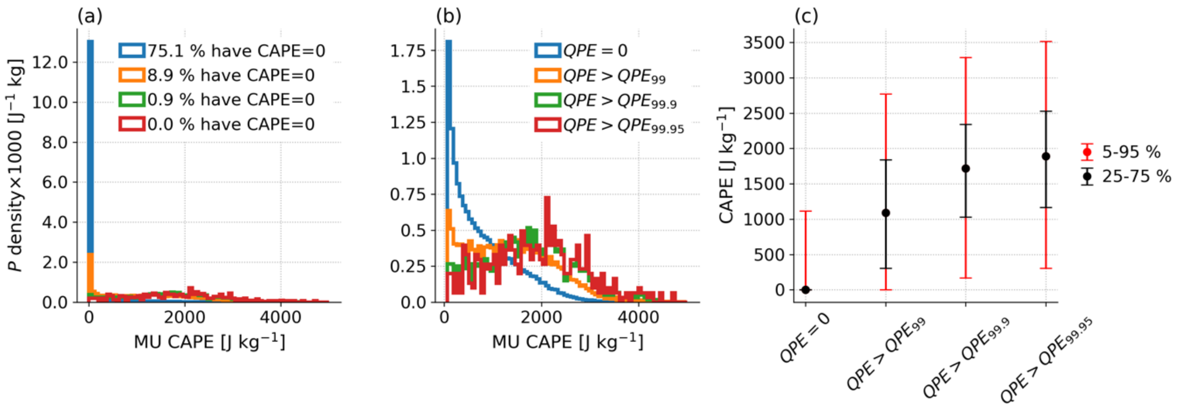


Supplementary Figure 8: Histogram of the area fraction of grid cells covered by convective (blue) or stratiform (orange) type precipitation flags according to the MRMS classification flags. (a) for cells where QPE is over its 95th percentile and (b) for cells where QPE exceeds its 99.95th percentile.

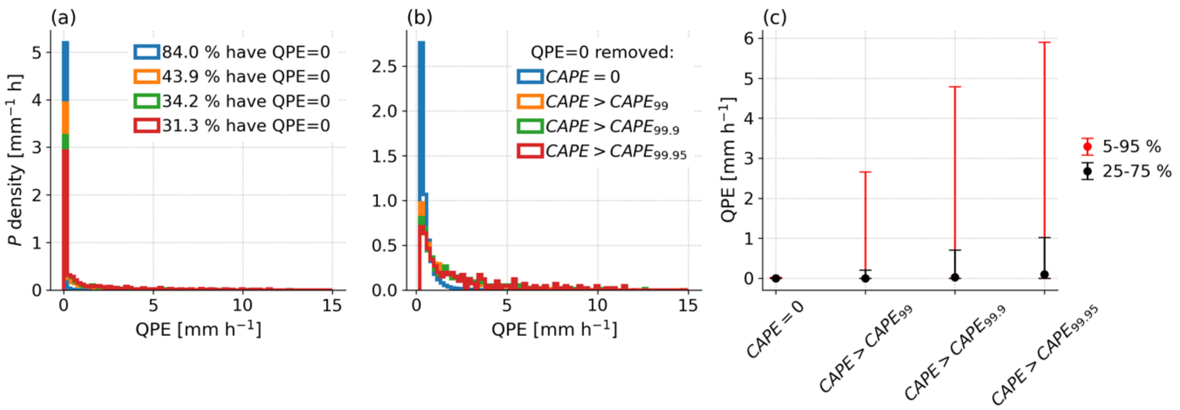
Supplementary Notes 6.

Supplementary Figure 9 shows how the CAPE distribution is shifted to higher values when more intense precipitation occurs. For example, when there is no precipitation, CAPE takes a value of 0 J kg^{-1} over 75 % of the time. In panel (c), there is a continuous shift upwards across all CAPE percentiles as we look at increasingly rare precipitation events.

Supplementary Figure 10 follows the same format and shows QPE when conditioned on CAPE. As expected, precipitation is more frequent and intense when CAPE is high, but in panel (c) the relationship is different. While heavy precipitation consistently goes along with high CAPE, it is not necessarily the case that high CAPE coincides with high precipitation. Rather, a far smaller fraction of high CAPE values results in intense precipitation, which can be seen from the increase in the 95th percentile of QPE in Supplementary Figure 10(c).



Supplementary Figure 9: Distributions of MU CAPE split by QPE thresholds. (a) histograms including all data, with percentage of poinst with CAPE=0 J kg^{-1} listed, (b) histogram with the zero valued bin removed, (c) median and percentile ranges of CAPE (interquartile or 5—95 %) in selected QPE bins.

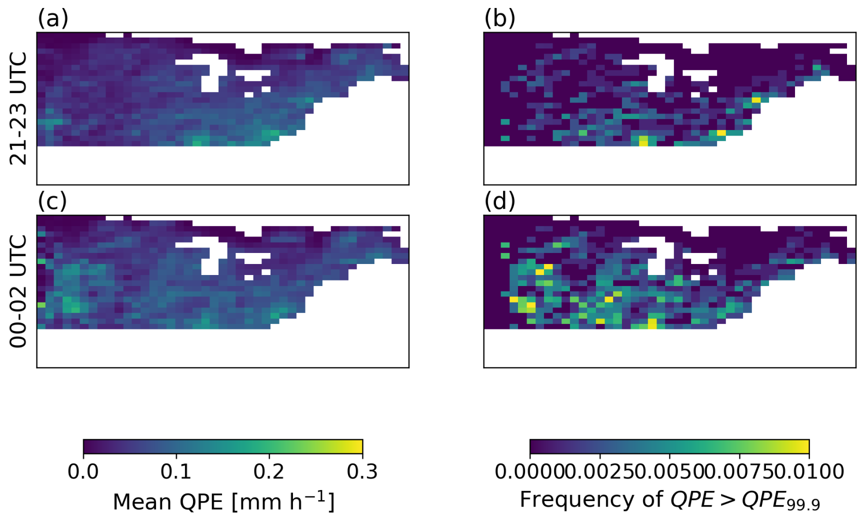


Supplementary Figure 10: Distributions of QPE split by MU CAPE thresholds. (a) histograms including all data, with percentage of points with QPE=0 mm h^{-1} listed, (b) histogram with the zero valued bin removed, (c) median and percentile ranges of QPE (interquartile or 5—95 %) in selected MU CAPE bins.

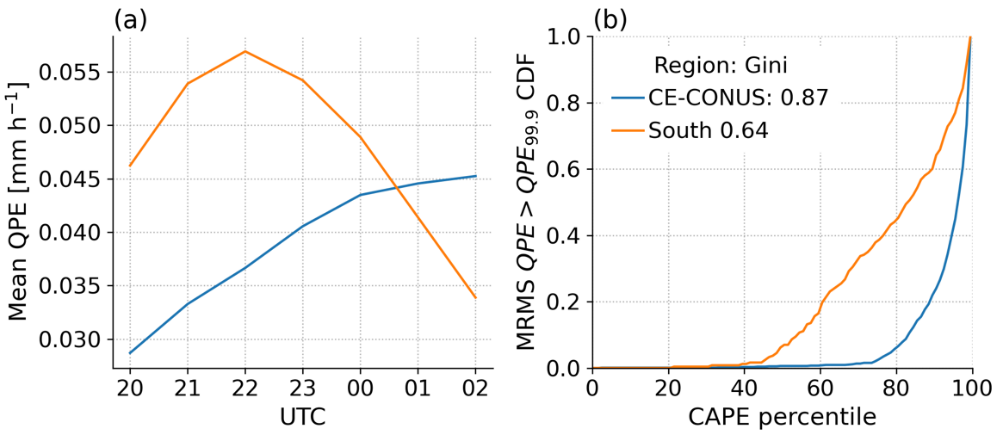
Supplementary Notes 7.

Supplementary Figure 11 shows that consistently heavy precipitation across the Plains and Midwest does not generally trigger until the later forecast hours 00—02 UTC. The earlier hours include more frequent events along the southernmost boundary, towards the Gulf of Mexico, and along the Atlantic coast.

We propose that poorer performance nearer the Gulf of Mexico shown in Supplementary Figure 12 may be related in part to the heavier ongoing precipitation that can be seen at AIRS overpass time, seen in Supplementary Figure 12(a). It is also likely that other processes will play a role, for example intense evaporation from the Gulf of Mexico likely moistens the offshore boundary layer and rises CAPE following the AIRS overpass to a level that is not typically seen inland.



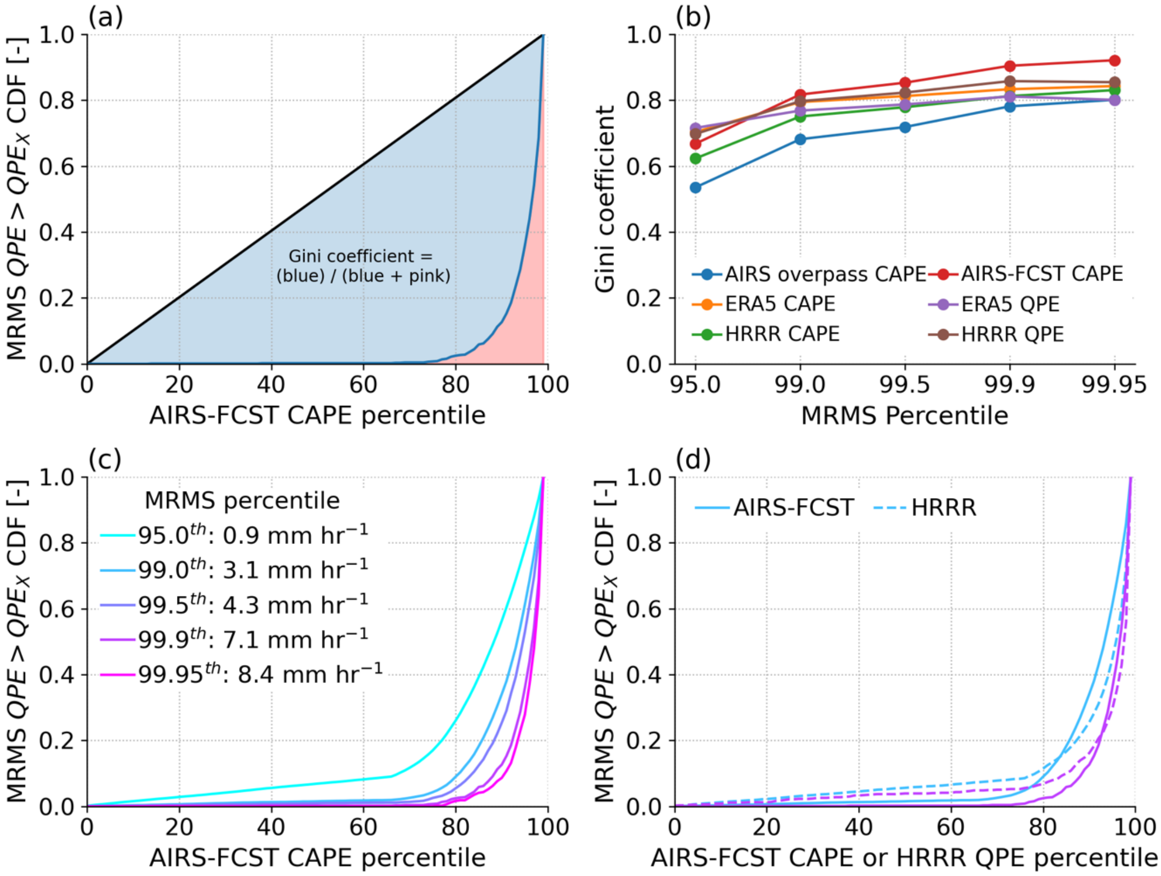
Supplementary Figure 11: (a) mean precipitation from 21—23 UTC, (b) frequency of QPE>QPE$_{99.9}$ from 21—23 UTC, (c,d) the same as above panels but for 00—02 UTC.



Supplementary Figure 12: (a) mean hourly precipitation and (b) CDF of QPE>QPE$_{99.9}$ for the studied CE-CONUS region (blue) and the land region south of 32 °N.

Supplementary Notes 8.

Most gridded datasets used in most studies of climate risk tend to report grid cell precipitation accumulation. A given accumulation could represent less-intense precipitation over a broad area, or more-intense precipitation over a smaller area. Each case is related to different hazards. We find that when Gini coefficients are calculated with reference to QPE calculated based on the precipitating area only, then Supplementary Figure 13 shows that grid-cell-mean CAPE is a superior predictor than for grid-cell mean prediction. We also show Gini coefficients skills obtained using HRRR and ERA5 grid-cell mean QPE, so it is unsurprising that their prediction skills are lower in this case. It is not trivial to convert HRRR and ERA5 output into a consistent "precipitating area mean" since they are at different resolutions to each other. Overall, it shouldn't be concluded here that CAPE is a better predictor than reanalysis or forecast model QPE, but rather that grid-cell mean CAPE is more informative about higher intensity but lower extent events, compared with lower intensity but higher extent events of the same rarity.



Supplementary Figure 13: as main manuscript Figure 3, but calculated using QPE intensity averaged over precipitating areas of the grid cell only.